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Application: The Costs of Taxation

PowerPoint Slides prepared by:
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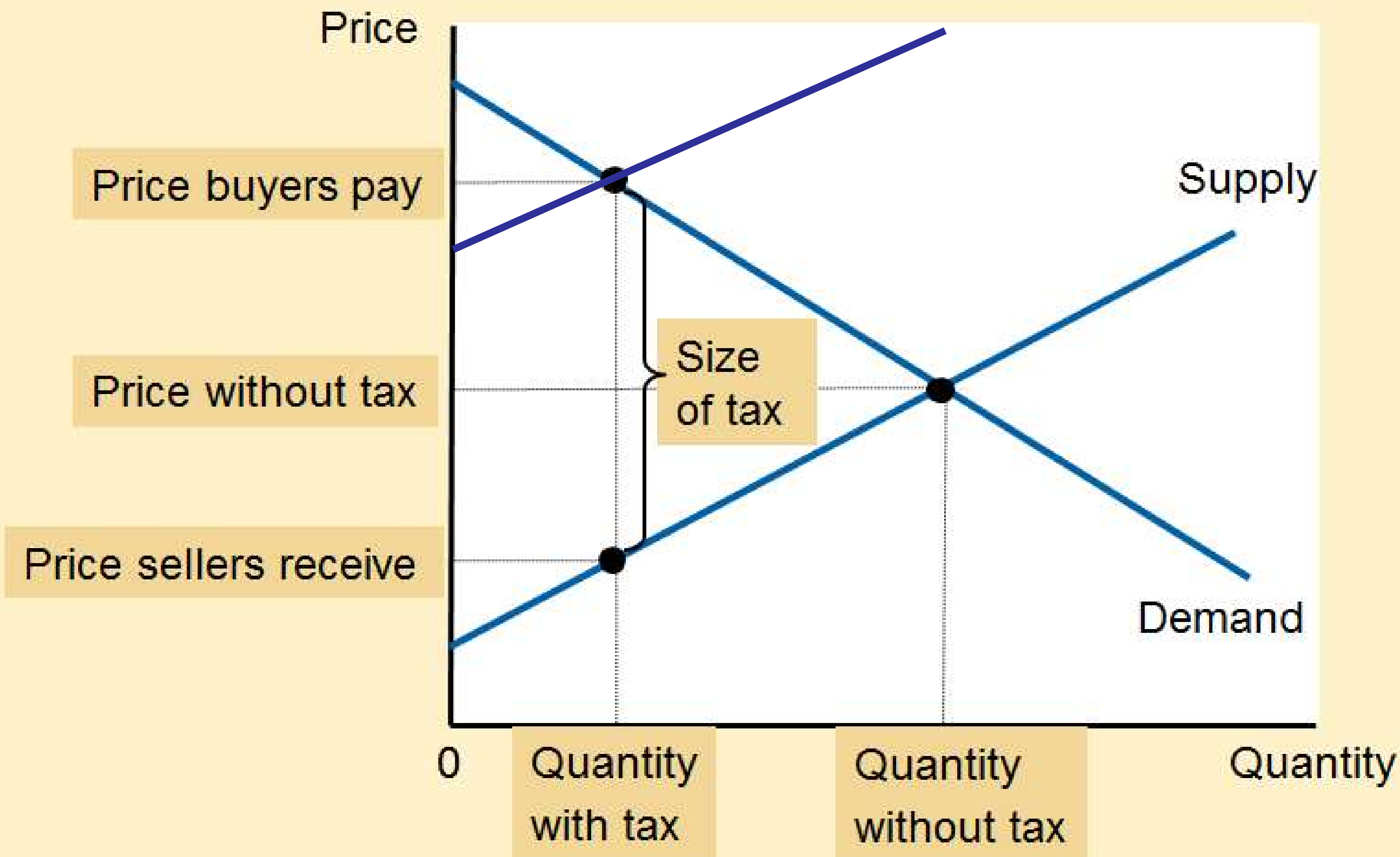
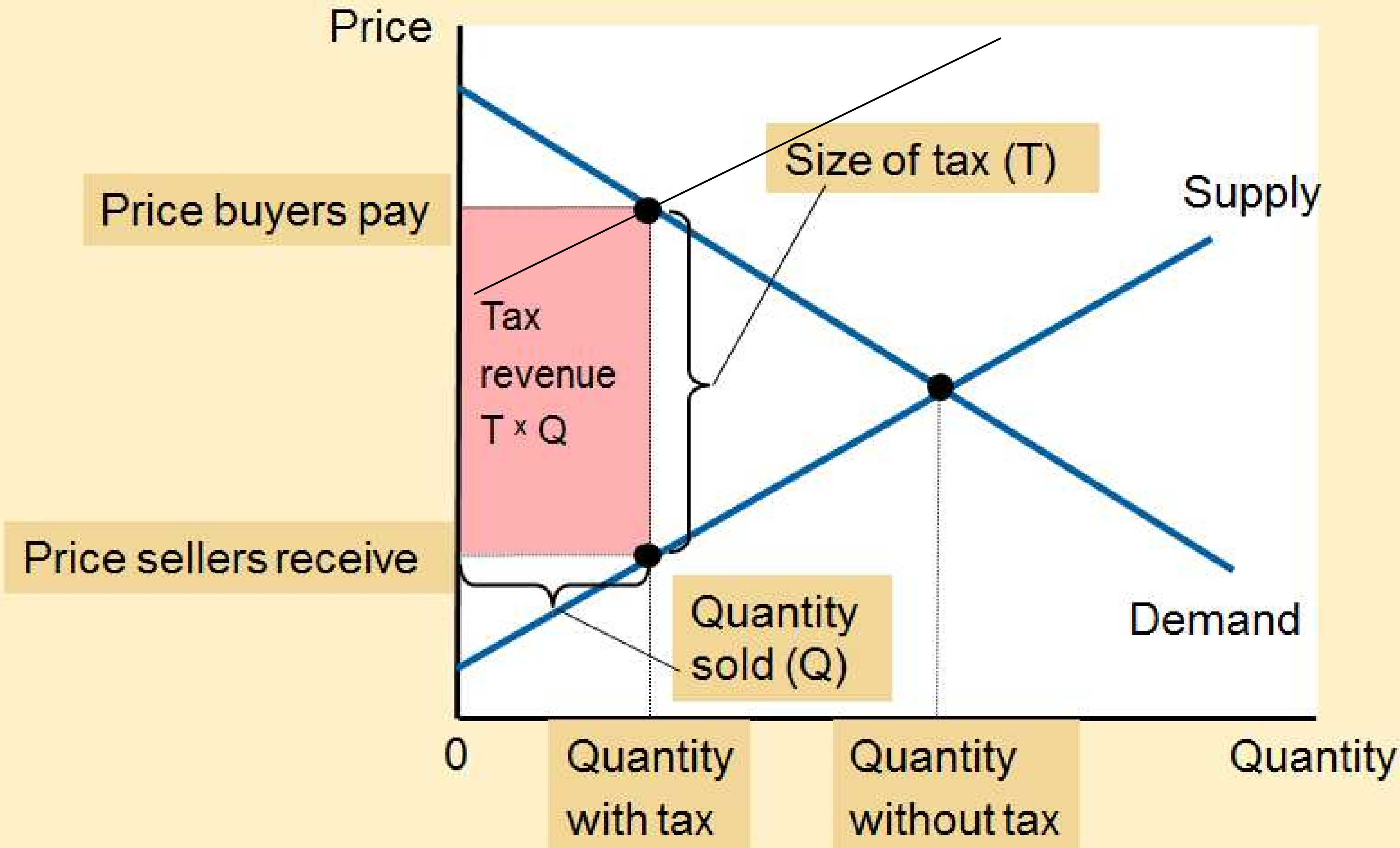
Figure 1

The Effects of a Tax



A tax on a good places a wedge between the price that buyers pay and the price that sellers receive. The quantity of the good sold falls.

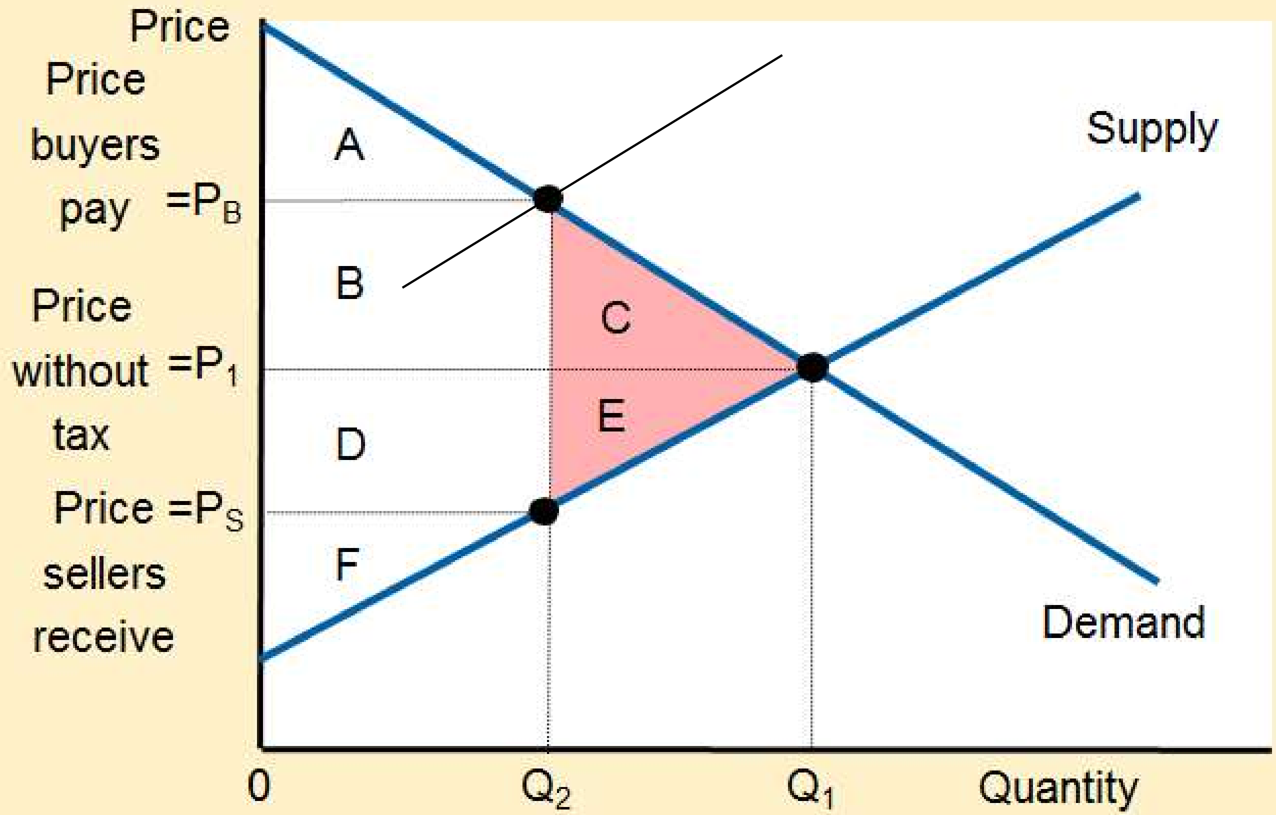
Figure 2
Tax Revenue



The tax revenue that the government collects equals T × Q, the size of the tax T times the quantity sold Q. Thus, tax revenue equals the area of the rectangle between the supply and demand curves.

Figure 3

How a Tax Affects Welfare



A tax on a good reduces consumer surplus (by the area B + C) and producer surplus (by the area D + E). Because the fall in producer and consumer surplus exceeds tax revenue (area B + D), the tax is said to impose a deadweight loss (area C + E).

	Without Tax	With Tax	Change
Consumer Surplus	A + B + C	A	-(B + C)
Producer Surplus	D +E + F	F	-(D + E)
Tax Revenue	None	B + D	+(B + D)
Total Surplus	A + B + C + D + E + F	A + B + D + F	-(C + E)

The area C + E shows the fall in total surplus and is the deadweight loss of the tax.

Deadweight Loss of Taxation, Part 5

- Welfare without a tax
  - Consumer surplus, areas A, B, and C
  - Producer surplus, areas D, E, and F
  - Total tax revenue = 0
- Welfare with tax
  - Smaller consumer surplus, area A
  - Smaller producer surplus, area F
  - Total tax revenue, areas B and D
  - Smaller overall welfare

Deadweight Loss of Taxation, Part 6

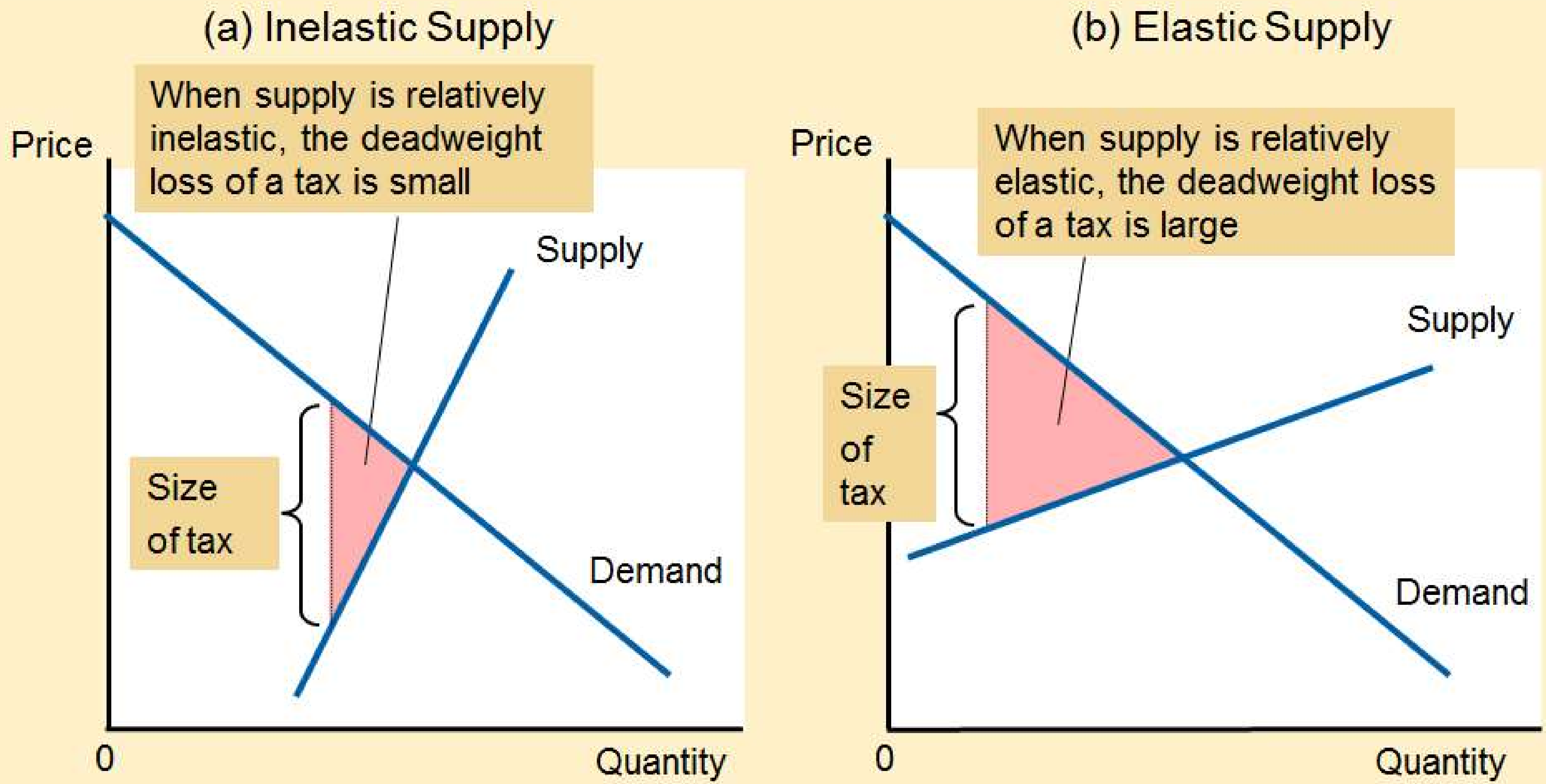
- Losses of surplus to buyers and sellers, from a tax
  - Exceed the revenue raised by the government
- Deadweight loss
  - Fall in total surplus that results from a market distortion, such as a tax
- Taxes distort incentives
  - Markets allocate resources inefficiently

Determinants of Deadweight Loss

- Price elasticities of supply and demand
  - More elastic supply curve
    - Larger deadweight loss
  - More elastic demand curve
    - Larger deadweight loss
- The greater the elasticities of supply and demand
  - The greater the deadweight loss of a tax

Figure 5

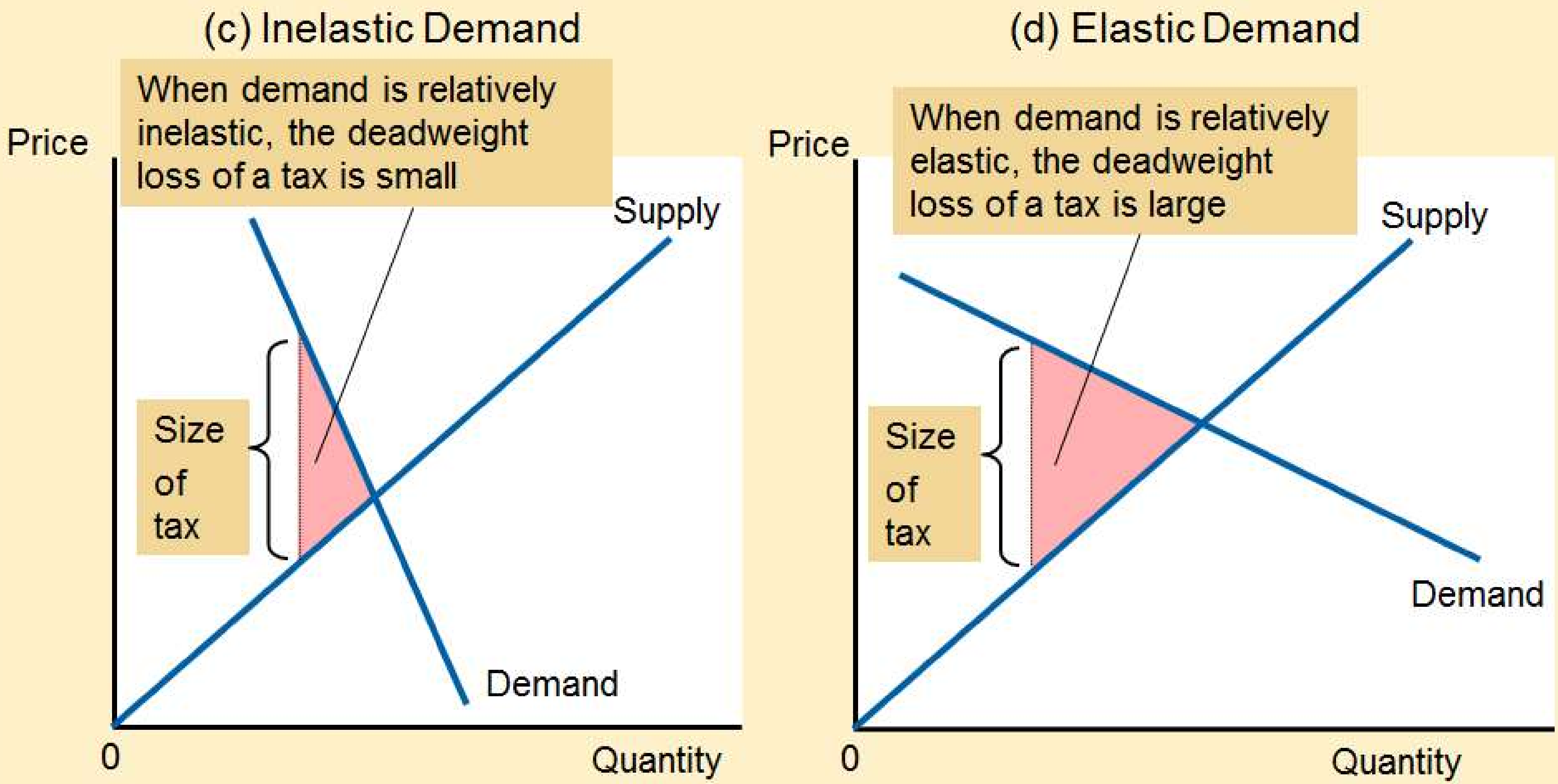
Tax Distortions and Elasticities (a, b)



In panels (a) and (b), the demand curve and the size of the tax are the same, but the price elasticity of supply is different. Notice that the more elastic the supply curve, the larger the deadweight loss of the tax.

Figure 5

Tax Distortions and Elasticities (c, d)



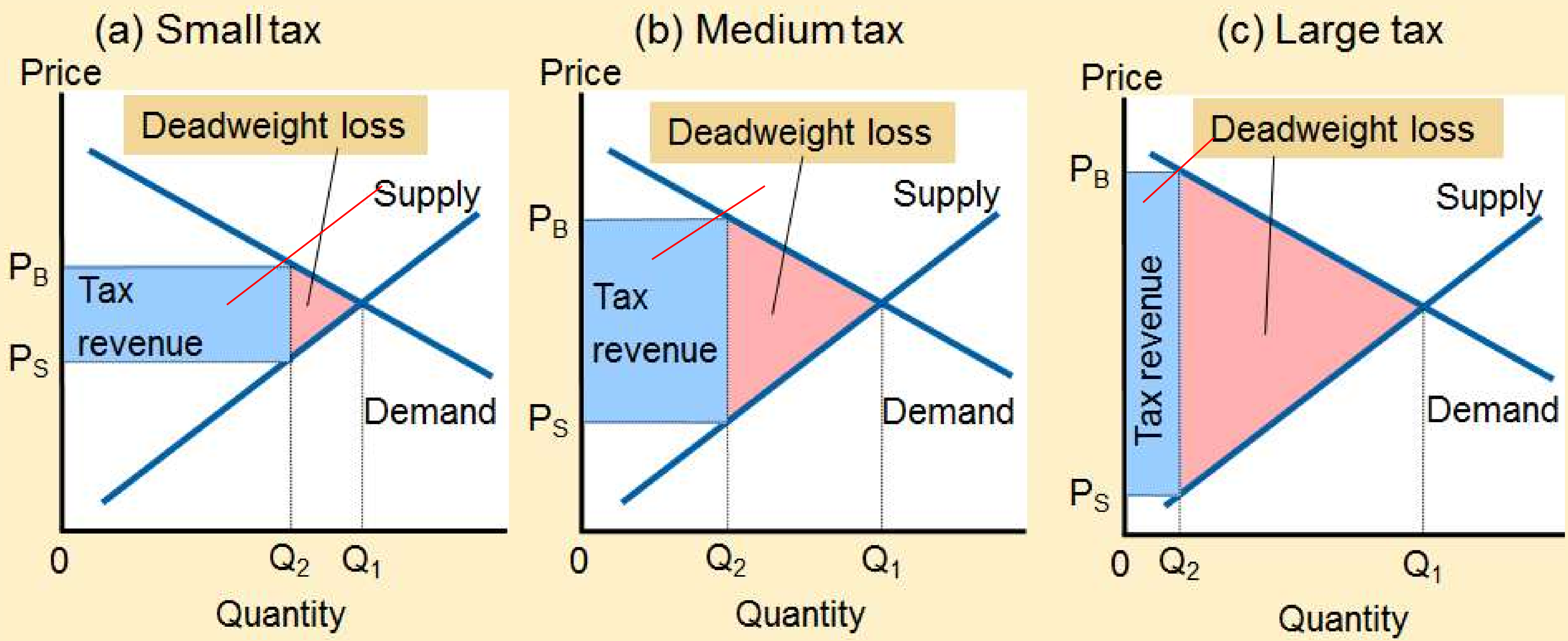
In panels (c) and (d), the supply curve and the size of the tax are the same, but the price elasticity of demand is different. Notice that the more elastic the demand curve, the larger the deadweight loss of the tax.

Deadweight Loss & Tax Revenue

- As the tax increases
  - Deadweight loss increases
    - Even more rapidly than the size of the tax
  - Tax revenue?

Figure 6

How Deadweight Loss and Tax Revenue Vary with the Size of a Tax (a, b, c)

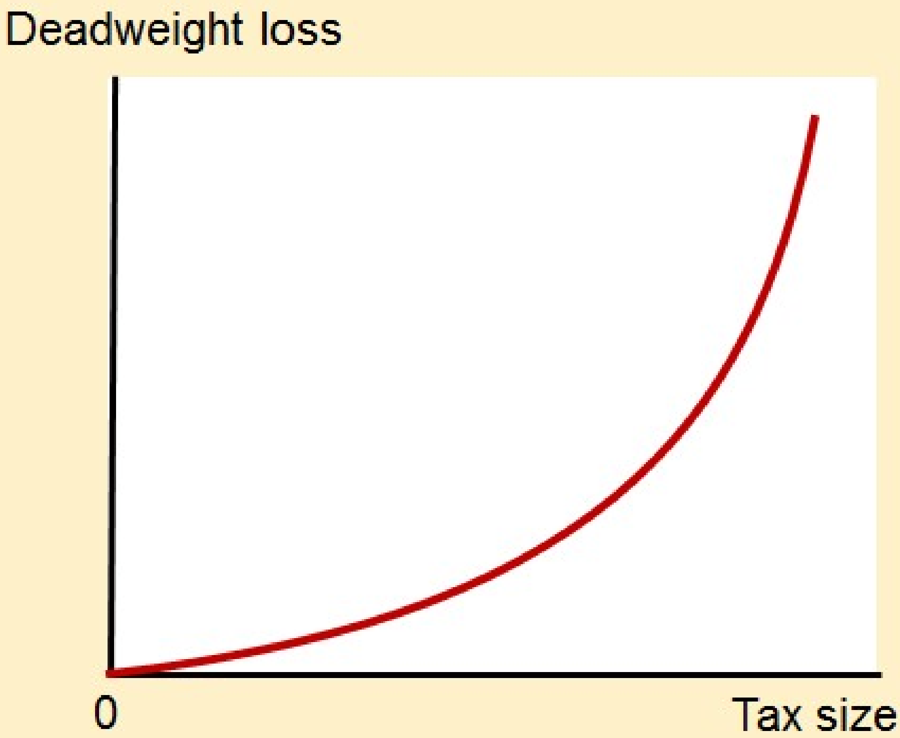


The deadweight loss is the reduction in total surplus due to the tax. Tax revenue is the amount of the tax times the amount of the good sold. In panel (a), a small tax has a small deadweight loss and raises a small amount of revenue. In panel (b), a somewhat larger tax has a larger deadweight loss and raises a larger amount of revenue. In panel (c), a very large tax has a very large deadweight loss, but because it has reduced the size of the market so much, the tax raises only a small amount of revenue.

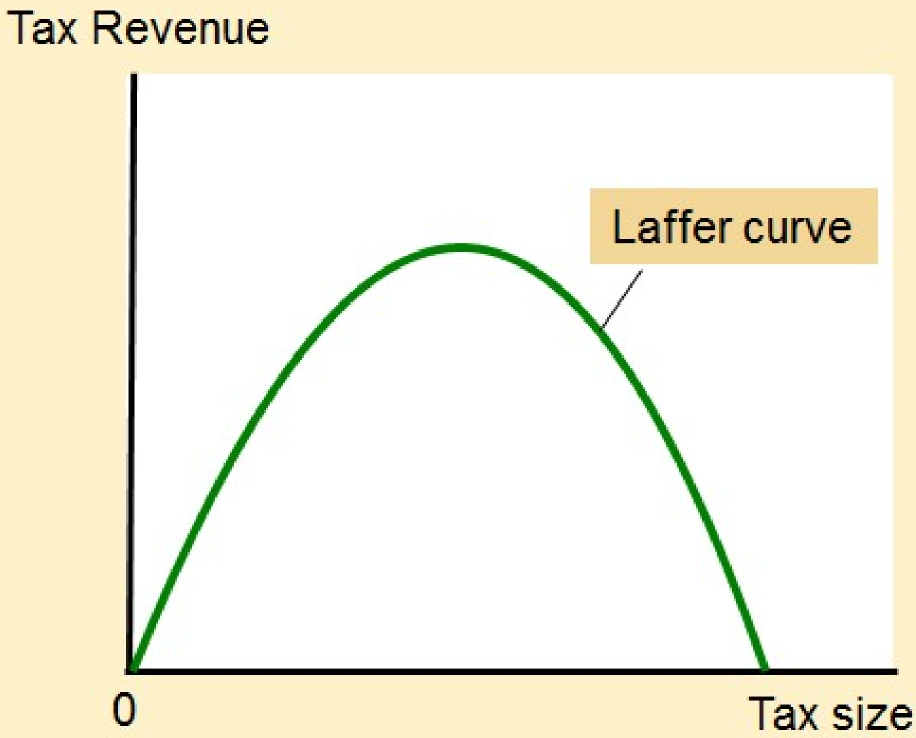
Figure 6

How Deadweight Loss and Tax Revenue Vary with the Size of a Tax (d, e)

(d) From panel (a) to panel (c), deadweight loss continually increases



(e) From panel (a) to panel (c), tax revenue first increases, then decreases



Panels (d) and (e) summarize these conclusions. Panel (d) shows that as the size of a tax grows larger, the deadweight loss grows larger. Panel (e) shows that tax revenue first rises and then falls. This relationship is sometimes called the Laffer curve.

The Laffer curve and supply-side economics, Part 1

- 1974, economist Arthur Laffer
  - Laffer curve
  - Supply-side economics
  - Tax rates were so high, that reducing them would actually raise tax revenue
- Ronald Reagan’s experience in film industry
  - High tax rates caused less work
  - Low tax rates caused more work

The Laffer curve and supply-side economics, Part 2

- Ronald Reagan ran for president in 1980
  - Platform: cutting taxes
  - Argument
    - Taxes were so high that they were discouraging hard work
    - Lower taxes would give people the proper incentive to work
  - Raise economic well-being
  - Perhaps increase tax revenue